Mostafa Habibi Dehsheikhi

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AI Engineer | Applied Scientist | Generative AI | Computer Vision | Multi-Agent Systems

PROFESSIONAL SUMMARY

AI Engineer and Ph.D. Candidate in Computer Engineering specializing in production-grade generative AI, multi-agent systems, retrieval-augmented generation (RAG), and computer vision. Experienced in architecting and deploying scalable AI applications using AWS Bedrock, Lambda, S3, OpenSearch, LangChain, and LangGraph, with expertise in Python, PyTorch, and end-to-end machine learning systems. Built intelligent AI solutions for healthcare, sports analytics, and enterprise applications, combining LLMs, computer vision, and cloud infrastructure. Author of 10 peer-reviewed publications, including an IEEE Best Paper Award.

TECHNICAL SKILLS

Programming Python, Java, C/C++, SQL, JavaScript, HTML/CSS, Bash
Generative AI LLMs, Multi-Agent Systems, Agentic AI, RAG, LangChain, LangGraph, Prompt Engineering, Tool Calling, Evaluation Pipelines, Structured Extraction
Machine Learning PyTorch, scikit-learn, Random Forest, SVM, Computer Vision, YOLO, MediaPipe, OpenCV, DeepFace
Cloud AWS Bedrock, Lambda, EC2, S3, API Gateway, OpenSearch, DynamoDB
Databases SQL, SQL Server, Database Design, Normalization, Data Modeling
Tools Git, Linux, Flask, MATLAB, n8n

PROFESSIONAL EXPERIENCE

Senior Scientist Intern, *AIMS Technologies* *May 2026 – Dec 2026*

- Architected and deployed production multi-agent AI systems for personalized running coaching, combining RAG, critic-agent validation, and iterative evaluation pipelines to improve recommendation accuracy and reliability.
- Built an AI-powered video biomechanics platform that analyzes running form from video and generates personalized technique recommendations using computer vision and LLM reasoning.
- Designed self-evaluation and critic-agent workflows to validate AI outputs before delivery, reducing hallucinations and improving consistency.
- Deployed the complete platform on AWS using Bedrock, Lambda, OpenSearch, DynamoDB, and S3 for scalable production inference.

Graduate Research Assistant, *University of Texas at Dallas, Richardson, TX* *Spring 2023 – Present*

- Led development of AI-powered healthcare monitoring systems for patient behavior analysis, posture recognition, activity recognition, and human-object interaction.
- Designed a scalable dual-camera computer vision platform using YOLO, stereo vision, triangulation, and DeepFace for behavioral monitoring.
- Built multi-agent LLM systems that convert unstructured clinical notes into FHIR-compliant structured data using LangChain, LangGraph, and open-weight LLMs.
- Developed vision-language pipelines combining computer vision and LLM summarization to automatically generate patient behavior reports.
- Developed AI systems for sports analytics, including posture classification, kinematic analysis, and athlete performance evaluation.
- Conducted research in low-light object detection, MRI radiogenomics, and cloud-based AI deployment using AWS.

Teaching Assistant, *University of Texas at Dallas, Richardson, TX* *5 Semesters*

- Assisted with *Introduction to Electrical and Computer Engineering* and *Introduction to Digital Systems* through lab support, grading, office hours, assignment guidance, and technical troubleshooting.
- Helped students understand core engineering and programming concepts, solve implementation issues, and complete coursework successfully.

Research Assistant, *Shahid Bahonar University, Kerman, Iran* *Aug 2013 – Aug 2015*

- Conducted research in computer science focused on mobile ad hoc networks (MANETs), routing algorithms, and intelligent systems.
- Fully funded by the Ministry of Science, Research, and Technology.

SELECTED ENGINEERING PROJECTS

- Developed a multi-agent Retrieval-Augmented Generation (RAG) platform using LangGraph, LangChain, ChromaDB, and local LLMs with specialized planning, retrieval, validation, and critique agents for domain-specific question answering.
- Built a serverless AWS application using Lambda, API Gateway, S3, and DynamoDB to provide scalable REST APIs and event-driven data processing.
- Developed a dual-camera human monitoring system using synchronized cameras, stereo triangulation, and deep learning to estimate 3D positions and monitor human activities.
- Built a real-time human interaction detection pipeline using YOLO, MediaPipe, and Random Forest to recognize person-object interactions and classify human postures from video streams.
- Designed and implemented a normalized relational database using EER modeling, SQL, stored procedures, triggers, indexing, and view optimization.
- Built workflow automation pipelines in n8n integrating REST APIs, databases, webhooks, and AI services to automate multi-step business processes.

AWARDS

- **Louis Beecherl, Jr. Graduate Fellowship** *2026–2027*
  Awarded a prestigious merit-based graduate fellowship by the Office of the Dean, Erik Jonsson School of Engineering and Computer Science, The University of Texas at Dallas, in recognition of outstanding academic and research excellence.
- **ECE Doctoral Excellence Award, The University of Texas at Dallas** *Spring 2026*
  Awarded in recognition of academic and research excellence by the Department of Electrical and Computer Engineering.
- **IEEE Best Paper Award, ICHI 2025** *June 2025*
  Awarded for the paper *Video-Based Human-Object Interaction Analysis for Patient Behavioral Monitoring* at the 13th IEEE International Conference on Healthcare Informatics, Rende, Italy.

EDUCATION

Ph.D. Candidate, Computer Engineering *Present*
The University of Texas at Dallas, Richardson, TX, USA GPA: 3.94/4.00
Dissertation: AI-powered video monitoring, computer vision, agentic AI, and healthcare analytics.
Relevant Coursework: Database Design, Virtual Reality, Machine Learning and Pattern Recognition, Computer Vision.

Master of Science, Computer Science *Aug 2015*
Shahid Bahonar University, Kerman, Iran GPA: 3.60/4.00
Thesis: *Bee-Hopnet Routing Algorithm in Mobile Ad Hoc Networks (MANETs).*
Relevant Coursework: Advanced AI, Image Processing, Machine Learning, Logic Programming, Issues in Smart Systems, Theory of Computer Sciences.

Bachelor of Science, Computer Science *Aug 2012*
Shahid Bahonar University, Kerman, Iran GPA: 3.46/4.00
Ranked 1st in the Computer Science Department.
Achieved 4th place in the ACM Programming Competition, Shahid Beheshti University, Tehran, 2011.
Project: Design, Development, and Implementation of a Finance and Human Resource System deployed in Java.

Selected Publications

1. **M. Habibi** and M. Nourani, "Agentic AI to Augment Unstructured Data for Information Exchange and LLM," *IEEE Journal of Biomedical and Health Informatics (JBHI)*, 2026. (Manuscript under review)
2. **M. Habibi**, M. Nourani, and D. H. Sullivan, "A Scalable Dual-Camera Platform for Behavioral Assessment in Healthcare," *2026 IEEE 14th International Conference on Healthcare Informatics (ICHI)*, Minneapolis, Minnesota, USA, 2026. (Institute for Healthcare Informatics (IHI) Student Award)
3. **M. Habibi** and M. Nourani, "An Integrated Deep Learning Architecture for Illumination-Aware Object Detection," *IEEE Southwest Symposium on Image Analysis and Interpretation (SSIAI)*, Santa Fe, New Mexico, USA, March 22-24, 2026.
4. **M. Habibi**, M. Nourani, and M. M. Nourani, "AI-Based Performance Analysis for Track and Field Athletes," in *47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Copenhagen, Denmark, July 14–17, 2025.
5. **M. Habibi**, Z. Delaram, M. Nourani, and D. H. Sullivan, "Video-Based Human-Object Interaction Analysis for Patient Behavioral Monitoring," in *2025 IEEE 13th International Conference on Healthcare Informatics (ICHI)*, Rende, Italy, 2025. (Best Paper Award)

Complete publication list available on Google Scholar.

CERTIFICATIONS

- **AI and Agentic Systems:** Understanding Prompt Engineering; Introduction to AI Agents; Building Scalable Agentic Systems; Multi-Agent Systems with LangGraph; Building AI Agents with Google ADK (DataCamp).
- **Generative AI:** Introduction to Generative AI Learning Path (Coursera / Google Cloud). AWS Certified AI Practitioner (Amazon)
- **Data Science:** IBM Data Science Professional Certificate; Introduction to Data Science (Coursera / IBM).
- **Technical and Vocational Certifications:** CCNA; Fundamentals of Robotics; Java.